Parametric (global) warping

- Examples of parametric warps:



translation

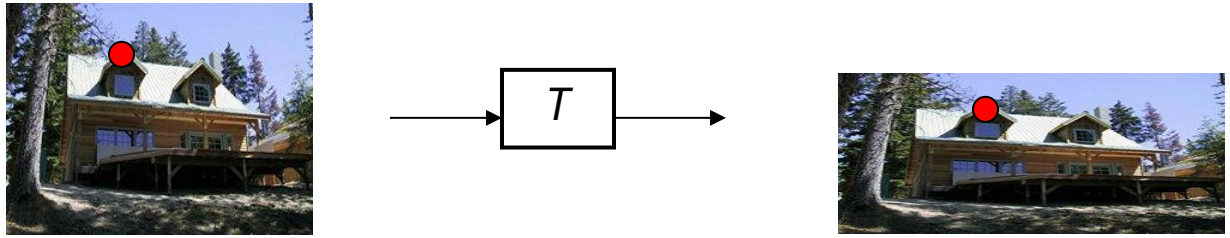


rotation



aspect

Parametric (global) warping



p = (x,y) **p′** = (x′,y′)

- Transformation T is a coordinate-changing machine:

$$\mathbf{p}' = T(\mathbf{p})$$

- What does it mean that *T* is global?
  - Is the same for any point **p**
  - can be described by just a few numbers (parameters)
- Let's consider *linear* xforms (can be represented by a 2x2 matrix):

$$\mathbf{p}' = \mathbf{T}\mathbf{p} \qquad \begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \end{bmatrix}$$

Common linear transformations

- Uniform scaling by *s*:

(0,0)



(0,0)



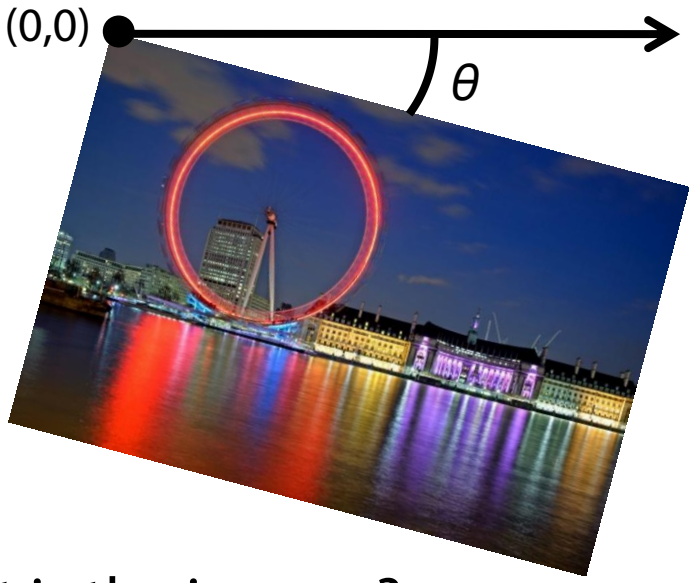
$$\mathbf{S} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix}$$

What is the inverse?

Common linear transformations

- Rotation by angle θ (about the origin)





$$\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

What is the inverse?

For rotations:

$$\mathbf{R}^{-1} = \mathbf{R}^{T}$$

2x2 Matrices

- What types of transformations can be represented with a 2x2 matrix?

  2D mirror across Y axis?

$$\begin{aligned} x' &= -x \\ y' &= y \end{aligned}$$

  2D mirror across line y = x?

$$\begin{aligned} x' &= y \\ y' &= x \end{aligned}$$

2x2 Matrices

- What types of transformations can be represented with a 2x2 matrix?

2D mirror across Y axis?

$$\begin{aligned} x' &= -x \\ y' &= y \end{aligned} \qquad \mathbf{T} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

2D mirror across line y = x?

$$\begin{aligned} x' &= y \\ y' &= x \end{aligned} \qquad \mathbf{T} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2x2 Matrices

- What types of transformations can be represented with a 2x2 matrix?

2D Translation?

$$
\begin{aligned}
x' &= x + t_x \\
y' &= y + t_y
\end{aligned}
$$

slido

Can a 2D translation by (tx,ty) be represented with a 2x2 matrix transformation?

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